Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/16/2007
PI Name:	Pillai, Suresh Ph.D.		
Project Title:	Microfluidic Device-Based Capture and Concentration of Microbial Contaminants in Recycled Water		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77845	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	01/02/2004	**End Date:**	04/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NCE to 4/30/2006 per PI (5/2006)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Beskok, Ali (Texas A& M University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	As the duration of space missions increase, recycling of metabolic and hygienic wastes becomes a necessity. Thus, the microbiological quality of recycled water is of critical importance to astronaut health and environmental quality during such missions. Accidental or catastrophic failures of the recycling process can introduce opportunistic organisms, pathogens and organisms that cause biofouling of water distribution lines. It is critically important that water quality monitoring systems are able to capture, concentrate and detect these contaminants effectively and efficiently. An efficient microbial capture and concentration device is needed for concentrating the sample prior to detection. The overall objective of this project is to design, fabricate, test, and optimize a prototype microfluidic device that can capture and concentrate microbial contaminants based on electrostatic trapping and electrophoretic motion principles. The microfluidic device will be designed to concentrate microbial contaminants from 120L/day of recycled water on a continuous basis. The salient feature of the device is that the final concentrated volume will be 1 mL which will		

	facilitate integration with any current or future contaminant detection technologies. In this pilot study proposal, we propose to demonstrate bacterial capture and concentration by the microfluidic device and detection by culture and PCR-based detection and characterization. This interdisciplinary project involving engineering and microbiology will delineate the design and performance characteristics of the microfluidic device. In addition to the benefits to NASA, the proposed technology has value in terrestrial and homeland security applications as well. The proposed project addresses the "Sample Acquisition and Handling" emphasis of the AEMC program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has provided valuable technical information about how to develop a microfluidic system to concentrate microorganisms from recycled water and drinking water. The technology developed in this project can be used by municipal water utilities to develop microbial monitoring programs
Task Progress:	Funding from the NASA Phase 1 Pilot Study project was used to design and demonstrate the applicability of a microfluidic device-based capture and concentration of microbial contaminants in recycled water. During the initial phases of this project we developed a basic understanding of the electrophoretic mobilities of selected bacteria under conditions that were realistic within recycled and drinking water distribution systems. Electrophoretic mobility and zeta potential measurements of bacterial cells in municipal drinking water can lead to enhanced understanding of bacterial surface interactions and adhesion to surfaces in realistic environmental conditions. During the second half of this project a large number of laboratory experiments were performed to design, fabricate and test the efficacy of a microfluidic-device based microbial capture system. These studies have led to the development of a microfluidic device based on electrophoretic transport and electrostatic trapping of charged particles for continuous capture and concentration of pathogens from water. Reclaimed and bottled water samples at pH values ranging from 5.2–6.5 were seeded with bacteria (E. coli, Salmonella, Pseudomonas) and viruses (MS-2 and Echovirus) and used in device testing. Both culture based methods and real-time PCR analysis were utilized to characterize the capture efficiency as a function of time, flow-rate, and applied electric field. Based on differences between the capture and negative control data, capture efficiencies of 90% to 99% were achieved for E. coli, Salmonella, Pseudomonas, and MS-2, while the capture efficiency for Echovirus was around 75%. Overall, the device exhibits 16.67 fold sample volume reduction within an hour at 6 mL/hr. This results in a concentration factor of 15 at 90% capture efficiency. Scaling of the device to sample 5 L/hr can be achieved by stacking 835 identical micro-channels. The device can thus function either as a filtration unit or as a sample concentrator to enable the application of real-time detection sensor technologies. The ability to continuously sample water without chemical additives facilitates the use of this device for NASA applications within enclosed environments. The successful completion of this project constitutes the first step in our development of a continuous, microbial capture and concentration system from large volumes of potable water. A key driving force in the device design was to achieve reduced mass, and reduce power and volume requirements.
Bibliography Type:	Description: (Last Updated: 03/19/2007)
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